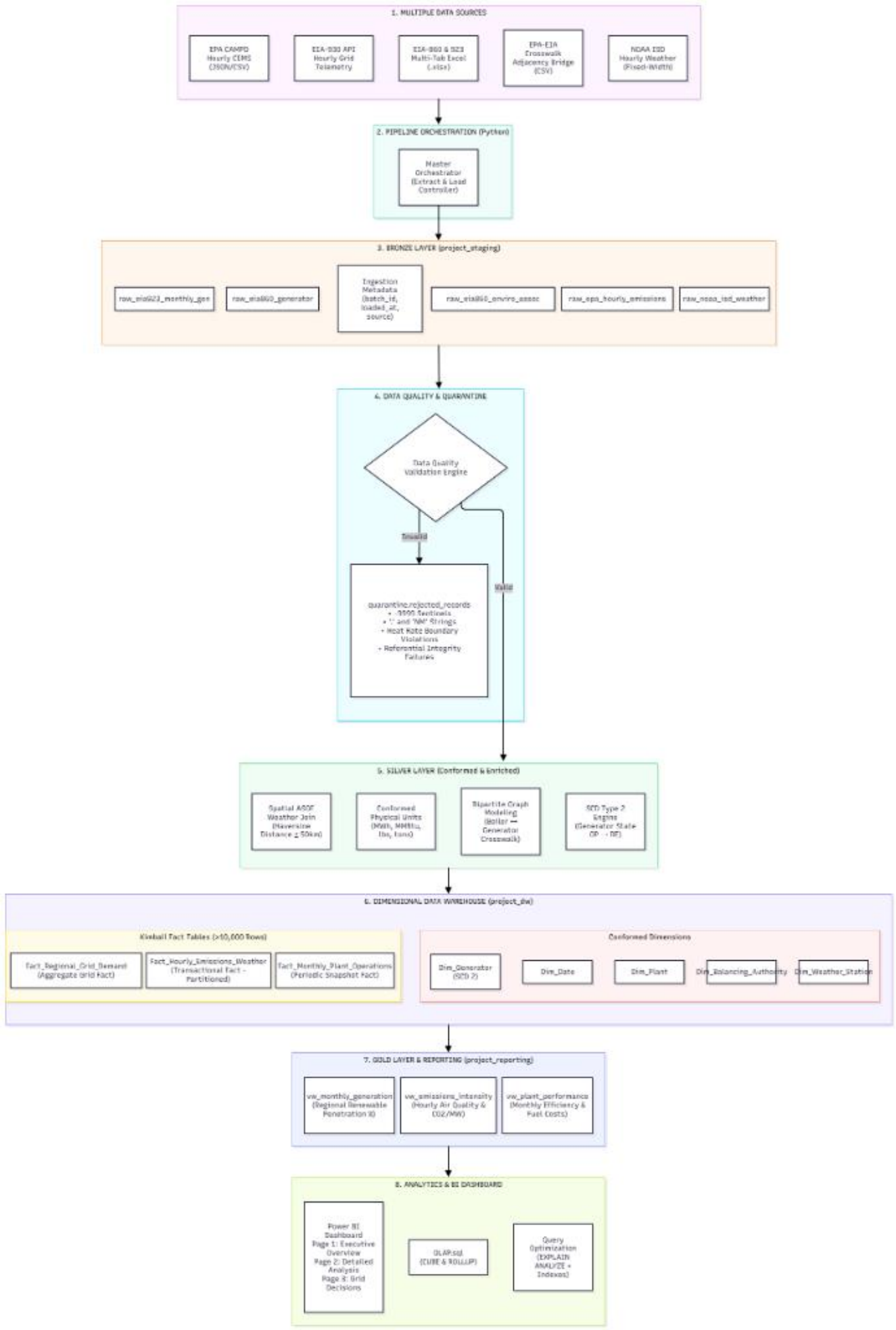
1. MULTIPLE DATA SOURCES
EPA CAMPD Hourly CEMS (350M/CSV)
EIA-930 API Hourly Grid Telemetry
EIA-860 & 923 Multi-Tab Excel (.xlsx)
EPA-EIA Crosswalk Adjacency Bridge (CSV)
NOAA ISD Hourly Weather (Fixed-Width)
2. PIPELINE ORCHESTRATION (Python)
Master Orchestrator (Extract & Load Controller)
3. BRONZE LAYER (project_staging)
raw_eia923_monthly_gen
raw_eia860_generator
Ingestion Metadata (batch_id, loaded_at, source)
raw_eia860_enviro_assoc
raw_epa_hourly_emissions
raw_noaa_isd_weather
4. DATA QUALITY & QUARANTINE
Data Quality Validation Engine
Invalid
Valid
quarantine.rejected_records
• -9999 Sentinels
• 'I' and 'NM' Strings
• Heat Rate Boundary Violations
• Referential Integrity Failures
5. SILVER LAYER (Conformed & Enriched)
Spatial ASOF Weather Join (Haversine Distance ≤ 50km)
Conformed Physical Units (MWh, MMBtu, lbs, tons)
Bipartite Graph Modeling (Boiler ↔ Generator Crosswalk)
SCD Type 2 Engine (Generator State OP ↔ RE)
6. DIMENSIONAL DATA WAREHOUSE (project_dw)
Kimball Fact Tables (>10,000 Rows)
Fact_Regional_Grid_Demand (Aggregate Grid Fact)
Fact_Hourly_Emissions_Weather (Transactional Fact - Partitioned)
Fact_Monthly_Plant_Operations (Periodic Snapshot Fact)
Conformed Dimensions
Dim_Generator (SCD 2)
Dim_Date
Dim_Plant
Dim_Balancing_Authority
Dim_Weather_Station
7. GOLD LAYER & REPORTING (project_reporting)
vw_monthly_generation (Regional Renewable Penetration %)
vw_emissions_intensity (Hourly Air Quality & CO2/MW)
vw_plant_performance (Monthly Efficiency & Fuel Costs)
8. ANALYTICS & BI DASHBOARD
Power BI Dashboard
Page 1: Executive Overview
Page 2: Detailed Analysis
Page 3: Grid Decisions
OLAP.sql (CUBE & ROLLUP)
Query Optimization (EXPLAIN ANALYZE + Indexes)